Khudhur Saeed

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Professional Summary

Computer engineering graduate candidate with a core dual focus in Backend Development and Cybersecurity. Combines expertise in designing secure, scalable backend architectures with hands-on application security and penetration testing experience gained through a 15-week internship. Skilled in building robust server-side systems, AI integration, and proactive vulnerability assessment, targeting roles in Software Engineering, Application Security, and DevSecOps.

Education

Bachelor of Science in Computer Engineering — Cumhuriyet Üniversitesi, Sivas

- **GPA:** 3.26 / 4.00
- **Status:** Graduated
- **7+1 Education System:** 7 academic semesters + 1 industry practice semester

Experience

Cybersecurity & Penetration Testing Intern — SETA Software Technologies Company (Feb 2026 – May 2026)

- Conducting penetration testing procedures within web application security
- Vulnerability detection and analysis based on OWASP Top 10 standards
- Manual testing of critical vulnerabilities such as IDOR, XSS, and SQL Injection
- HTTP traffic inspection and security vulnerability scanning using Burp Suite
- Network mapping, information gathering (Reconnaissance), and service enumeration processes
- Port scanning and service version analysis using Nmap
- Performing exploitation phases using the Metasploit Framework
- Identifying misconfigurations and weaknesses in Active Directory environments
- Simulating lateral movement techniques using Pass-the-Hash and Kerberoasting methods
- Applying privilege escalation (PrivEsc) techniques on Windows and Linux systems
- Conducting vulnerability assessment and reporting security findings
- Solving hands-on lab environments on TryHackMe and Hack The Box platforms
- Applied standardized Penetration Testing Methodology: Reconnaissance → Enumeration → Exploitation → Post-Exploitation → Reporting

Technical Skills

- **Programming Languages:** Python, Go (Golang), Java
- **Backend & Systems:** FastAPI, RESTful APIs
- **AI & LLM:** Google Gemini API, RAG Architectures
- **Databases & Search Engines:** PostgreSQL, SQL Server, Elasticsearch, Chroma Vector DB,

SQLAlchemy (Async), Alembic
- **Applied Cryptography:** AES-256-GCM, Argon2id, Zero-Knowledge Architecture, Challenge-Response Auth
- **DevOps & Infrastructure:** Docker, Docker Compose, Git, Linux, Bash Scripting
- **Threat Analysis & Security:** OWASP Top 10, Burp Suite, Nmap, Metasploit, Active Directory, Privilege Escalation

Key Projects

Cross-Platform Remote Monitoring & Endpoint Detection Platform (RMM / EDR) | GitHub Repository

Tech Stack: Go (Golang), Python, FastAPI, React, WebSockets, PostgreSQL, Redis, Docker, Cryptography (Ed25519, SHA-256)

- **High-Performance Cross-Platform Agent (Go):** Engineered a low-footprint background agent for Linux (systemd) and Windows (Windows Services), utilizing native OS syscalls, /proc filesystem, and Win32 APIs for zero-dependency hardware telemetry (CPU, RAM, disk, network sockets, active processes).
- **Secure C2 & Real-Time Communications:** Architected an authenticated bidirectional Command & Control (C2) channel via WebSockets (RFC-6455) with TLS enforcement, connection heartbeat watchdogs, Redis-backed state synchronization, and exponential backoff with decorrelated jitter.
- **Cryptographic Task Signing & Integrity:** Implemented an end-to-end task execution pipeline where remote scripts are signed on the server using Ed25519 and verified by agents before execution, mitigating tampering and unauthorized command injection.
- **Endpoint Detection & Auditing (EDR):** Built native security monitors including File Integrity Monitoring (FIM) with SHA-256 hashing, Linux authentication log tracking (auth.log/secure), and native Windows Event Log streaming for critical security IDs (logons, privilege escalation).
- **Interactive Remote Shell & Automation:** Integrated an interactive web-based terminal streaming PTY/ConPTY sessions to the browser using xterm.js, along with automated unattended deployment scripts (PowerShell & Bash) and single-use enrollment tokens.
- **Robust Backend & Dashboard:** Developed asynchronous REST APIs and WebSocket endpoints using FastAPI, SQLAlchemy (Async), and PostgreSQL, featuring RBAC, immutable audit logging, threshold-based alerting engines, and a responsive React monitoring dashboard.

SecureBox (Zero-Knowledge Password Manager) | GitHub Repository

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy 2.0 (Async), Alembic, Docker, Typer, Rich, Argon2id, AES-256-GCM

- **Zero-Knowledge Architecture:** Engineered an end-to-end Zero-Knowledge password manager architecture where cryptographic keys, master passwords, and plaintext records are never exposed to or stored on the backend server.
- **Key Derivation Engine:** Implemented client-side dual-key derivation using Argon2id, bifurcating output into an ephemeral in-memory Master Encryption Key (MEK) and an authentication verification key (MAK).
- **Authenticated Encryption:** Developed local authenticated symmetric encryption using AES-256-GCM with unique 12-byte CSPRNG nonces and record UUIDs bound as Associated Data (AD) to prevent ciphertext tampering and record-swapping attacks.
- **Envelope Encryption Scheme:** Designed an envelope encryption scheme with a Symmetric Vault Key (SVK) wrapped via MEK, enabling O(1) master password rotation without re-encrypting vault items and supporting offline emergency recovery keys.
- **Authentication Handshake:** Implemented an anti-enumeration, HMAC-based two-phase challenge-response authentication handshake with replay-resistant ephemeral nonces and short-lived JWT session issuance.

- **Asynchronous REST APIs:** Built asynchronous RESTful APIs using FastAPI and SQLAlchemy (asyncpg), enforcing strict binary-level constraints (BYTEA) and schema migrations via Alembic.
- **Interactive CLI & CSPRNG:** Created an interactive CLI interface with local in-memory vault search, secure key memory purging upon exit, and an integrated CSPRNG password generator with Shannon entropy calculation.

Clothing Store E-Commerce Platform | GitHub Repository

Tech Stack: Python, FastAPI, SQLAlchemy, PostgreSQL, Elasticsearch, Google Gemini, Docker

- **Async Backend Architecture:** Developed a scalable asynchronous backend architecture using FastAPI and managed database migrations with Alembic.
- **Semantic Search System:** Implemented a product-based semantic search system integrating Elasticsearch 8.13.
- **AI Integration & Features:** Integrated three AI-powered features using Google Gemini: AI Chatbot, Virtual Try-On, and Outfit Builder.
- **Authentication & RBAC:** Enforced secure authentication using JWT and bcrypt, alongside Role-Based Access Control (RBAC) for authorization.
- **API Documentation & Containerization:** Built 20+ RESTful API endpoints documented via Swagger/OpenAPI and containerized the system using Docker Compose.

PDF Chat Application (RAG) | GitHub Repository

Tech Stack: React.js, TypeScript, FastAPI, Python, LangChain, Google Generative AI (Gemini), Chroma, Tailwind CSS

- **Async API Architecture:** Built a scalable and asynchronous backend architecture using FastAPI
- **Security & CORS Configuration:** Minimized cross-origin security risks through secure CORS configuration
- **Global Exception Handling:** Designed a centralized and secure API response structure using global exception handling mechanisms
- **Vector RAG Architecture:** Implemented an embedding-based RAG architecture using Chroma vector database
- **Document Isolation & Access Control:** Ensured data security and access control through multi-PDF upload and document isolation system
- **Session-Based Chat:** Developed session-based chat architecture to provide persistent and user-specific interactions
- **Document Analysis Pipeline:** Designed an async pipeline for high-performance document analysis.

Languages

- **Arabic:** Native
- **Turkish:** Advanced (C1)

English: Intermediate (B1–B2)